



Fig. 4: Micro-supercapacitor

ultra-high power handling with an R-C time constant of only 26ms.

Onion-like carbon

Although OLC has been synthesised by many different methods in the last 30 years, large-scale production (gram quantities) of OLC was first realised in 1994 using vacuum annealing of a nanodiamond precursor. Annealing in inert gases to transform nanodiamond (currently produced in large quantities) to OLC is also used by many researchers. This method has a potential for industrial applications, as the onion yield is close to 100 per cent and the manufacturing volume is only limited by the size of the furnace, and can be scaled accordingly. This material rarely has ideal spherical carbon onions, but can be produced in large quantities for practical applications.

Another synthesis technique using arc discharge between two graphite electrodes in water produces OLC of slightly different structure than from annealing of nanodiamond. A DC current of 30A and 17V applied between two graphite electrodes in water causes the carbon to evaporate

at the location of the arc due to the extreme heat generated. The carbon vapour rapidly condenses into highly spherical OLC particles.

Hollow carbon onions have been produced with the help of metal nanoparticles. One practical method for the fabrication of hollow carbon onions uses nitric acid to dissolve nickel from the carbon-coated nickel nanoparticles. In this method, the decomposition of methane in the presence of Ni/Al catalyst particles produces carbon nano-onions as the primary product.

Carbon-encapsulated metal (magnetic) nanoparticles represent a new class of zero-dimensional carbon-metal composite nanomaterials. These materials are very easily purified to the form of hollow onions by the subsequent nitric acid treatment.

There are several other processes to produce carbon onions. Synthesis of carbon onions via chemical vapour deposition (CVD) utilises an iron catalyst supported on sodium chloride to decompose acetylene gas at 400°C temperature. Today, catalytic chemical vapour deposition is considered as the only economi-

cally viable process for large-scale carbon nanotube production and their integration into various devices. In contrast to other methods of carbon onion synthesis, this CVD process yields much larger-diameter (50nm) particles.

Carbon ion implantation is another method to produce carbon onions, which allows the particle diameter to be tuned from 3nm up to 30nm by varying synthesis conditions such as temperature or implantation dose density. Thermolysis, in which a compound is decomposed by heat, has been shown to be a method for carbon onion synthesis. Using sodium azide (NaN_3) and hexachlorobenzene (C_6Cl_6) as reagents, a redox reaction causes an abrupt increase in temperature and pressure, producing large-diameter carbon nanotubes.

Recent progress in on-chip micro-supercapacitors

A novel method is being developed for fabricating micro-patterned interdigitated electrodes based on reduced graphene oxide (rGO) and carbon nanotube (CNT) composites for ultra-high-power-handling micro-supercapacitor applications.

There are two main approaches to fabrication of devices with interdigital electrodes. In the first approach, interdigital current collectors are fabricated and then the active materials deposited onto the current collectors using thin-film deposition method. In the second approach, a patterning step is performed on a thin film of active material to form the interdigital electrodes. **EFY**

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